

A student team building an autonomous boat from the ground up.

One composite planing hull, two motor pods steering by differential thrust, and a rudder for stability at speed. Designed as one system and built by students at UW-Madison.



WHAT MAKES US DIFFERENT

Designed as one system

One optimization model picks hull shape, drive, battery and speed together, so no part is sized alone.

Industry experience

Members have worked at Tesla, Northrop Grumman, Xcel Energy, Milwaukee Tool, John Deere and Fincantieri.

ENGINEERS FROM ACROSS UW-MADISON

UW-Madison trains engineers in every discipline. We pulled from each segment of the boat.

Industrial

Optimization, composites and manufacturing

Mechanical

Structures, CAD and fabrication

Electrical

Power, battery and high voltage

Computer

Embedded, firmware and software

OUR APPROACH

Hull

Plane well, carry the load, stay light.

- Hybrid planing composite hull
- Shape tuned by optimization
- Resistance checked in CFD first

Electric

Spend every watt-hour on speed.

- High-voltage battery storage
- Motor pods sized by the model
- Differential thrust and rudder

Autonomous

Find the way and steer, with no one at the helm.

- GPS, IMU and LIDAR sensing
- Firmware protects the battery
- Software plans the route

TIMELINE

Oct 25
Hull design

Nov 15
Electrical design

Nov 25
Hull built

Jan 25
First drive

Feb 15
First fast drive

Mar 15
Tuning and testing

Sponsor the build

We are looking for funds, parts and advisors.

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CONTACT

Marley Barrett, Project lead
mhbarrett@wisc.edu